

project summary
phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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The atomization and evaporation of fuel sprays is a fundamental process that takes place in internal combustion engines. This proposal focuses on new numerical methods for carrying out “digital twin” atomization experiments, leading to the improvement in the efficiency and/or performance of jet engines and rotating detonation rocket engines.

Effective Numerical methods and experiments alike must reliably measure multiscale processes ranging from millimeters to nanometers. Macro scale sprays break up into sheets and ligaments, which in turn break up into a mist. Evaporation turns the mist into a gaseous fuel which is ready to be reacted with oxidizer. The spray breakup process, along with convective heat transfer from evaporation and preceding reactions, induces a feedback mechanism in which the resulting turbulent flow field affects the breakup. At the same time, for rotating detonation engines, not all of the droplets need to be atomized prior to shock wave interaction. Remaining micro-scale sized droplets that survived the “pre-shock” atomization process are shattered by the supersonic detonation wave into a sub-micron “fog” almost instantly. At which point reactions occur, thereby generating the detonation wave for the next cycle. It is noted that in an optimized rotating detonation engine, the “pre-shock” remaining droplets must not be too large (so as to survive the shock wave interaction) and not too small either (so as not to have used too much energy in the pre-shock atomization phase).

It is with this backdrop that the reality is that both high-fidelity atomization and spray simulations and physical experiments are very expensive. This proposal’s objective is to develop novel Artificial Intelligence (AI) methods for making the best use of data that is available. The proposed work consists of 3 parts: (1) data assimilation, (2) AI learning of multiphase flow sub-scale models, and (3) verification and validation.

1. Recent multi-phase flow data assimilation algorithms have been developed by Dommermuth et. al. for assimilating measured sea surface data for the purpose of improving incompressible multiphase flow predictions as well as enabling the discovery of hidden experimental states. We shall develop analogous data assimilation techniques with the extended capability for compressible multiphase flows.
2. The discrepancy between a coarse grid simulation, with data assimilation, and the corresponding high fidelity data, is used as training data for a neural network model in order to learn the coarse grid correction term, using solely coarse grid simulation data.
3. Coarse grid simulations are carried out for atomization and spray cases not included in the original data assimilation process (step 2). The coarse grid simulations are carried out with/without the aid of the neural network model developed in case 2 above.